

Preliminary Experiment

Filippo Spinelli

August 2026

1 Preliminary problem

We will denote the point in three-dimensional space as $\mathbf{y} = (y_1, y_2, y_3)$, where $y_i, i = 1, 2, 3$, are the spatial coordinates. Suppose we have a fixed conductivity field $k : \Omega \rightarrow \mathbb{R}$, where the domain $\Omega = [0, 1]^2 \times [0, \frac{1}{2}]$ represents the three-dimensional integrated circuit. The field consists of two layers, varies only along the y_2 axis, and is constant in the other two dimensions; in particular

$$k(\mathbf{y}) = \begin{cases} 1.4 \text{ [W/(mK)]} & \text{if } y_2 < \frac{10}{21}, \\ 14/19 \text{ [W/(mK)]} & \text{if } y_2 = \frac{10}{21}, \\ 0.5 \text{ [W/(mK)]} & \text{if } y_2 > \frac{10}{21}, \end{cases} \quad (1)$$

where $\frac{14}{19}$ is the harmonic mean between the values $k_1 = 1.4$ and $k_2 = 0.5$; indeed, all points with $y_2 = 10/21$ represent the interface between the two materials.

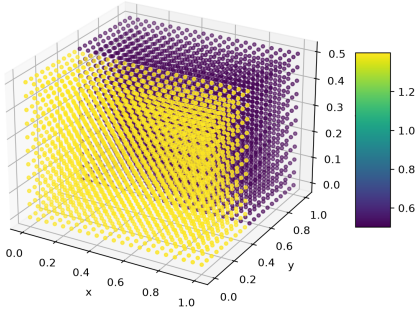


Figura 1: 2 layers conductivity map

We will work on a fixed grid, which discretizes our domain into $21 \times 21 \times 11$ points. The ground-truth solution was generated with **Ansys Electronic Student Desktop 2025**, where the layer transition occurs exactly at $y_2 = \frac{10}{21}$ so that the grid has the first 9 slices in the first region, slice number 10 represents the interface between the two materials, and the next 11 slices belong to the

second region. The power map was constructed so as to heat both conductivity regions and is given by a rectangular region where the imposed heat flux is constant and equal to $2500 [W/m^2]$; on the rest of the top face, the flux is imposed to be constantly equal to 0. It was then rescaled following the same normalization as in the article.

Normalization of the power map

The power distribution is defined on a subdivision of 20×20 tiles on the upper surface of the cube. To each tile of the power map, a dimensional power is associated by multiplying the corresponding dimensionless value by the scaling factor $0.00625 \text{ mW} = 6.25 \times 10^{-6} \text{ W}$.

Let p_{ij}^* be the dimensionless value associated with tile (i, j) , with $i, j = 1, \dots, 20$. The corresponding dimensional power is therefore

$$P_{ij} = p_{ij}^* 6.25 \times 10^{-6} \text{ W}.$$

Now consider the case in which a uniform heat flux equal to $q = 2500 \text{ W/m}^2$ is imposed on the upper surface.

Denoting by A_{top} the total area of the upper surface, the area of each tile is

$$A_{\text{tile}} = \frac{A_{\text{top}}}{20 \times 20} = \frac{A_{\text{top}}}{400}.$$

The dimensional power associated with each tile in the presence of a uniform flux is therefore

$$P_{ij} = q A_{\text{tile}} = q \frac{A_{\text{top}}}{400}.$$

By equating this expression with the scaling relation adopted for the power map, one obtains the corresponding dimensionless value $p_{ij}^* = \frac{q A_{\text{top}}}{400 \cdot 6.25 \times 10^{-6}}$.

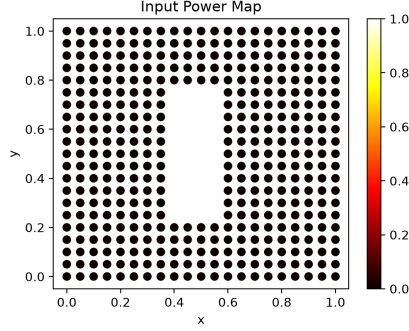
In the present case, the area of the upper surface is assumed to be $A_{\text{top}} = 1 \text{ mm}^2 = 10^{-6} \text{ m}^2$.

It follows that the area of each tile is $A_{\text{tile}} = \frac{10^{-6}}{400} = 2.5 \times 10^{-9} \text{ m}^2$, and therefore the power associated with each tile is

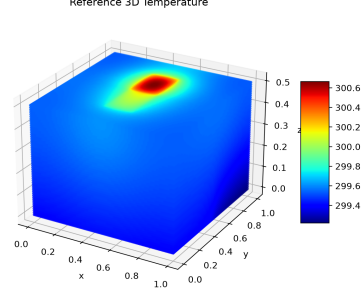
$$P_{ij} = 2500 \cdot 2.5 \times 10^{-9} = 6.25 \times 10^{-6} \text{ W}.$$

This value coincides exactly with the adopted scaling factor. Consequently, the corresponding dimensionless value of the power map is equal to one, namely $p_{ij}^* = 1$. We thus conclude that a constant physical heat flux of 2500 W/m^2 corresponds, in the adopted dimensionless representation, to a uniform power map with unit value on all tiles, that is,

$$p_{ij}^* = 1, \quad \forall i, j = 1, \dots, 20.$$



(a) Rescaled power map



(b) True temperature

In figure (a), the rescaled power map according to the article's convention is shown; at the black points, the heat flux is zero, while in the white region a constant flux of $2500 [W/m^2]$ is imposed. In figure (b), the true solution of the problem generated with **Ansys Electronic Student Desktop 2025** is shown and represented on a uniform grid of $101 \times 101 \times 51$ points.

Loss function

The loss function used is structured as follows:

$$\mathcal{L}_{pde}(\theta) = \frac{1}{N_{pde}} \sum_{i=1}^{N_{pde}} |k(\mathbf{y}_i) \Delta T(\mathbf{y}_i)|^2,$$

where $\{\mathbf{y}_i\}_{i=1}^{N_{pde}} \subset \Omega \setminus (\partial\Omega \cup [0, 1] \times \{\frac{10}{21}\} \times [0, \frac{1}{2}])$ are the grid points that do not belong either to a face of the cube or to the interface.

$$\mathcal{L}_{side}(\theta) = \frac{1}{N_{side}} \sum_{j=1}^{N_{side}} \left| k(\mathbf{y}_j) \frac{\partial T}{\partial n}(\mathbf{y}_j) \right|^2,$$

where $\{\mathbf{y}_j\}_{j=1}^{N_{side}} \subset \partial\Omega \setminus ([0, 1] \times [0, 1] \times \{0, \frac{1}{2}\})$ are the grid points that belong to the lateral walls of the cube.

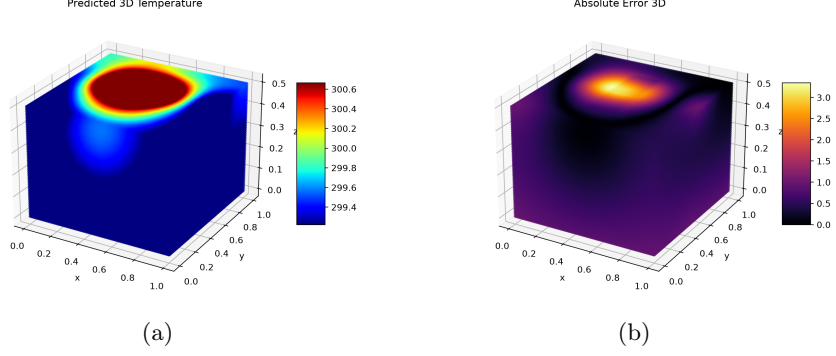
$$\mathcal{L}_{top}(\theta) = \frac{1}{N_{top}} \sum_{s=1}^{N_{top}} \left| k(\mathbf{y}_s) \frac{\partial T}{\partial n}(\mathbf{y}_s) - q(\mathbf{y}_s) \right|^2,$$

where $\{\mathbf{y}_s\}_{s=1}^{N_{top}} \subset [0, 1] \times [0, 1] \times \{\frac{1}{2}\}$ are the grid points that belong to the upper face of the cube.

$$\mathcal{L}_{bottom}(\theta) = \frac{1}{N_{bottom}} \sum_{r=1}^{N_{bottom}} \left| k(\mathbf{y}_r) \frac{\partial T}{\partial n}(\mathbf{y}_r) - h(T(\mathbf{y}_r) - T_{amb}) \right|^2,$$

where $\{\mathbf{y}_r\}_{r=1}^{N_{bottom}} \subset [0, 1] \times [0, 1] \times \{0\}$ are the grid points that belong to the lower face of the cube. The final loss function will be of the form:

$$\mathcal{L}_{tot}(\theta) = \lambda_{pde}\mathcal{L}_{pde}(\theta) + \lambda_{side}\mathcal{L}_{side}(\theta) + \lambda_{top}\mathcal{L}_{top}(\theta) + \lambda_{bottom}\mathcal{L}_{bottom}(\theta) \quad (2)$$



In figure (a), the prediction of the temperature field obtained with our model is shown. In figure (b), the absolute error between the true temperature and the prediction made with DeepOHeat-v1-k is shown

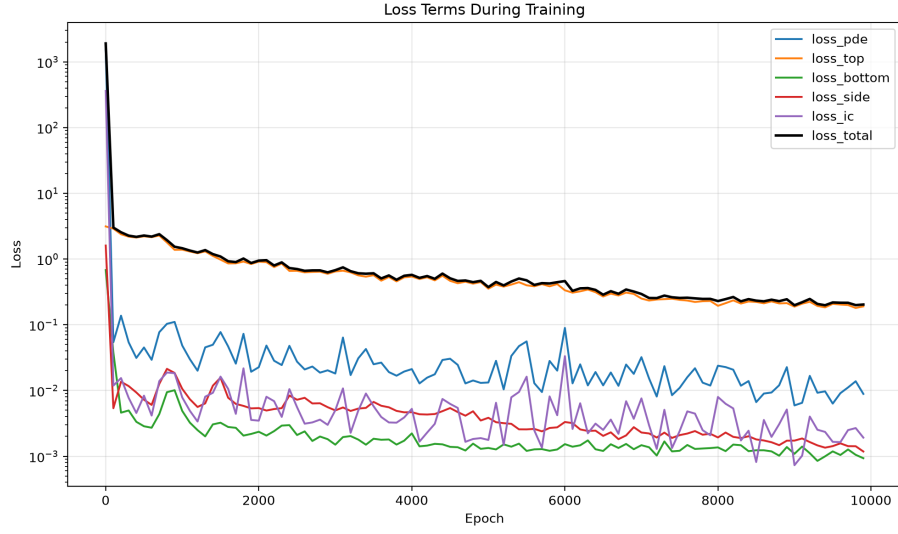


Figure 4: Plot of all the terms that make up the final loss function (Loss_ic has been separated from the rest of Loss_pde). As can be seen from the figure, the only problem lies in the loss function on the top of the cube, in making the model learn the power map.

Modification 1: Doubling the iterations during training

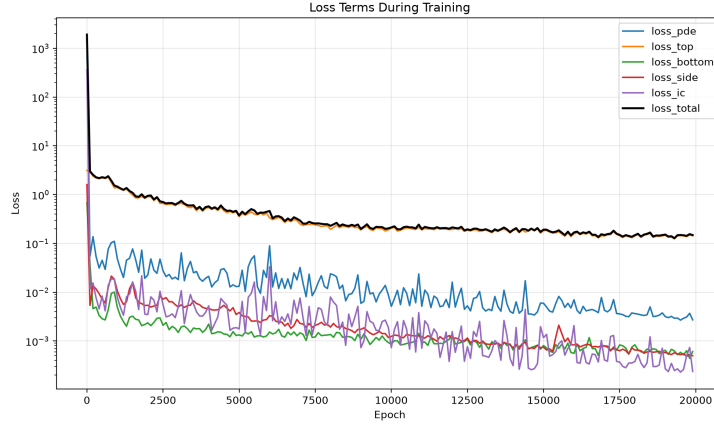
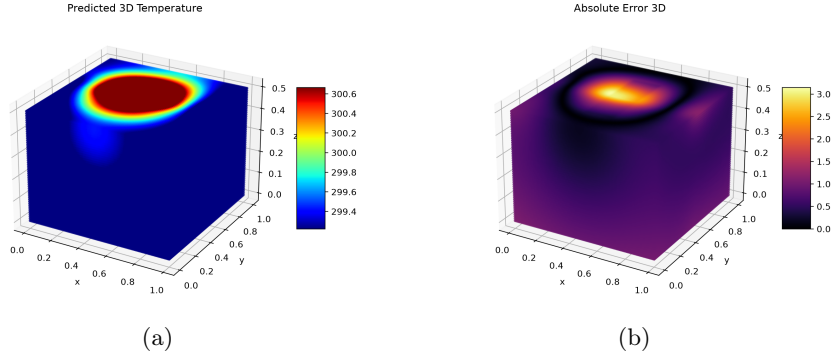


Figure 5: Trend of the terms that make up the loss function over the 20000 training iterations

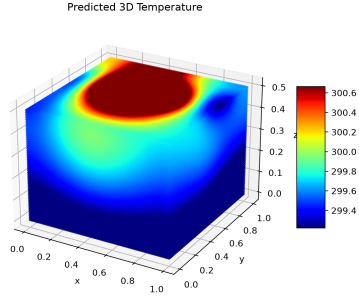


Prediction (a) and absolute error (b) for the case with 20000 training iterations

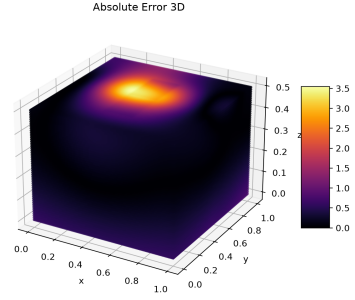
Modification 2: Batch change

The training of the model works as follows. At each iteration:

- 50 random conductivity maps are sampled among the 10000 available
- We input all 50 conductivity maps into the branch networks
- At each point, the loss function is computed as the average of the loss functions across all the configurations in the iteration.

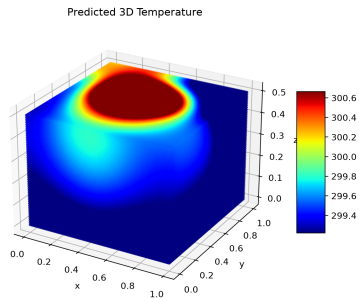


(a)

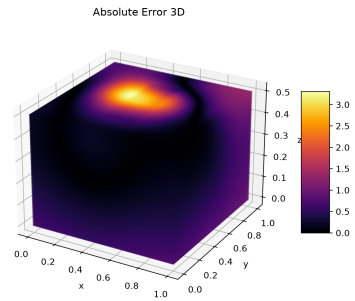


(b)

Prediction (a) and absolute error (b) with 30 maps per iteration

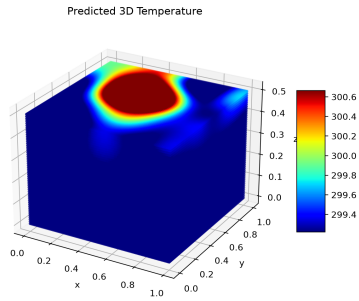


(a)

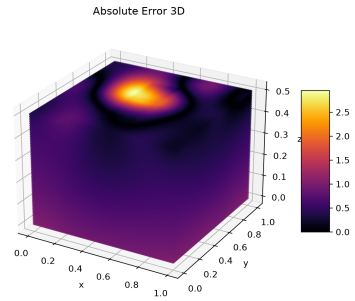


(b)

Prediction (a) and absolute error (b) with 70 maps per iteration



(a)



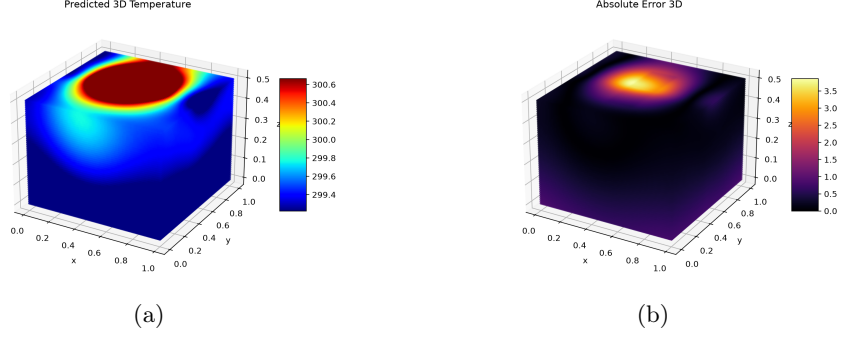
(b)

Prediction (a) and absolute error (b) with 300 maps per iteration

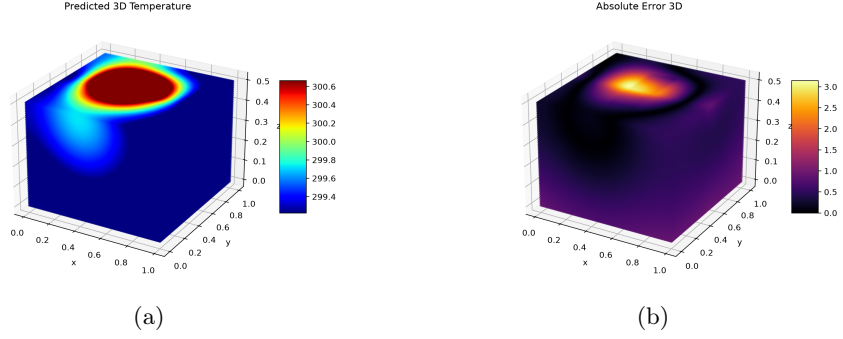
Modification 3: Weight of the Loss_top

$$\mathcal{L}_{tot}(\theta) = \lambda_{pde}\mathcal{L}_{pde}(\theta) + \lambda_{side}\mathcal{L}_{side}(\theta) + \lambda_{top}\mathcal{L}_{top}(\theta) + \lambda_{bottom}\mathcal{L}_{bottom}(\theta) \quad (3)$$

We try changing the parameter λ_{top} in (3) to give more weight to estimation errors when approximating the power map on the upper face. By combining

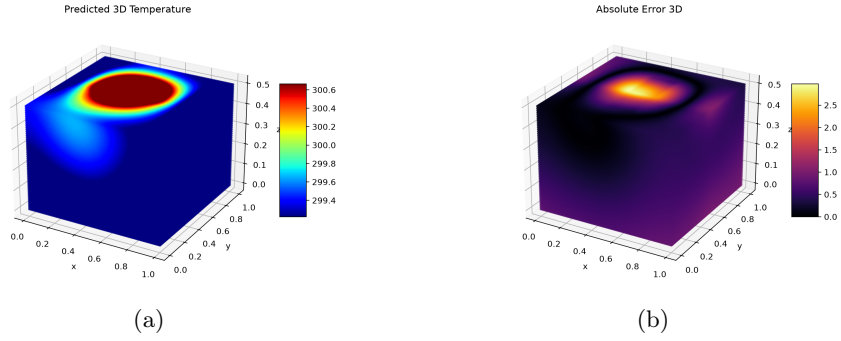


Prediction (a) and absolute error (b) with $\lambda_{top} = 2$

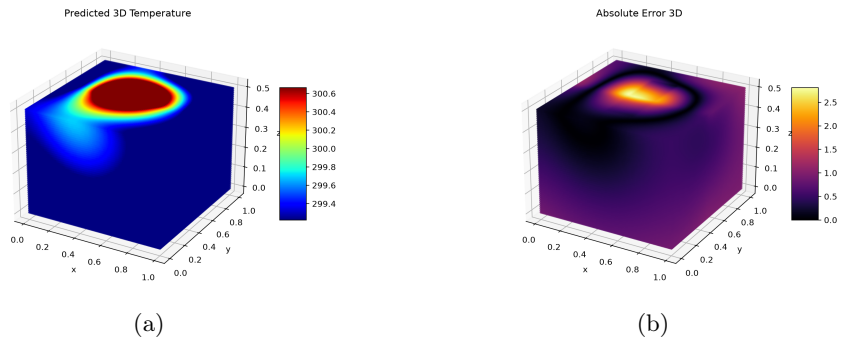


Prediction (a) and absolute error (b) with $\lambda_{top} = 4$

the conditions $\lambda_{top} = 16$ and $N_{iter} = 20000$, the following results are obtained:



Prediction (a) and absolute error (b) with $\lambda_{top} = 8$



Prediction (a) and absolute error (b) with $\lambda_{top} = 16$

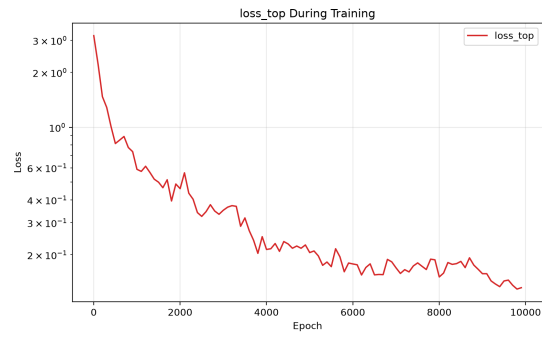
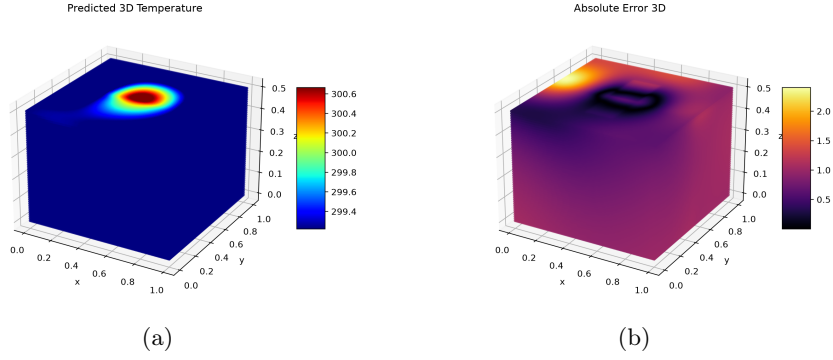


Figure 14: Trend of the top loss with $\lambda_{top} = 16$



Prediction (a) and absolute error (b) with $\lambda_{top} = 32$ and 20000 training iterations

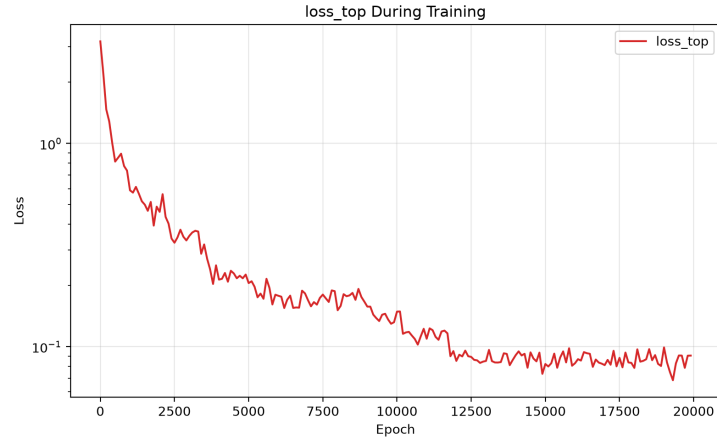
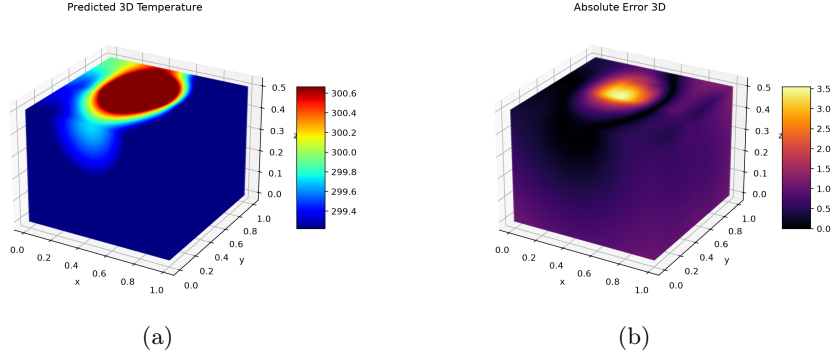


Figura 16: Trend of the top loss with $\lambda_{top} = 16$ for 20000 training iterations



Prediction (a) and absolute error (b) with $\lambda_{top} = 16$ and 20000 training iterations

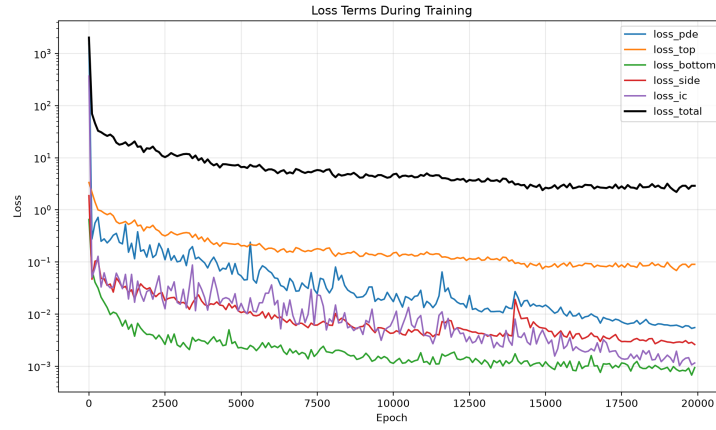
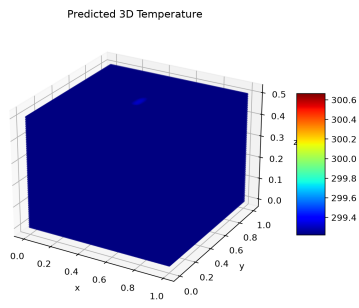


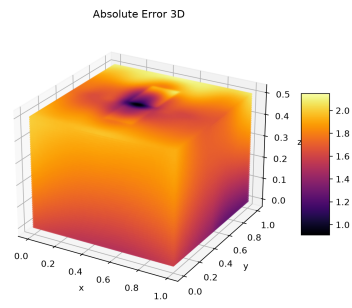
Figura 18: Trend of all terms of loss function with $\lambda_{top} = 32$ for 20000 training iterations

Modifica 4: Cambio del training set

Fino ad adesso gli esperimenti sono stati tutti condotti con lo stesso training set dell'articolo degli autori, dato da 10000 campi gaussiani che generano mappe liscie 21×21 . Vediamo cosa succede se addestriamo la rete con un training set dato da 5000 campi gaussiani e 5000 mappe a blocchi con potenze scalate che arrivano fino a 4. I risultati che otteniamo con 25000 iterazioni, 1000 decay steps e $\lambda_{top} = 1$ sono i seguenti.



(a)



(b)

Prediction (a) and absolute error (b) with $\lambda_{top} = 1$ and 25000 training epochs